

Examples on Probability

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Ex. A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the prob. that among the balls drawn there is at least one ball of each colour.

Solⁿ: The required event E that 'in a draw of 4 balls from the box at random there is at least one ball of each colour', can materialise in the following mutually disjoint ways:

- i) 1 red, 1 white, 2 black balls
- ii) 2 red, 1 white, 1 black balls
- iii) 1 red, 2 white, 1 black balls

$$\begin{aligned} P(E) &= P(i) + P(ii) + P(iii) \\ &= \frac{{}^6C_1 \times {}^4C_1 \times {}^5C_2}{{}^{15}C_4} + \frac{{}^6C_2 \times {}^4C_1 \times {}^5C_1}{{}^{15}C_4} + \frac{{}^6C_1 \times {}^4C_2 \times {}^5C_1}{{}^{15}C_4} \\ &= 0.5275. \end{aligned}$$

Ex. A problem in Statistics is given to the three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$ respectively. What is the prob. that the problem will be solved if all of them try independently.

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{3}{4}, \quad P(C) = \frac{1}{4}$$

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ &\quad + P(A \cap B \cap C) \\ &= P(A) + P(B) + P(C) - P(A)P(B) - P(A)P(C) - P(B)P(C) \\ &\quad + P(A)P(B)P(C) \\ &= \frac{29}{32}. \end{aligned}$$

$E_1 \rightarrow$ manufactured by A $P(E_1) = 0.25$

$E_2 \rightarrow$ " B $P(E_2) = 0.35$

$E_3 \rightarrow$ " C $P(E_3) = 0.40$

Hence,

$$P(E_1|D) = \frac{P(E_1)P(D|E_1)}{\sum_{i=1}^3 P(E_i)P(D|E_i)}$$
$$= \frac{0.25 \times 0.05}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02}$$
$$= \frac{25}{69}$$
$$P(E_3|D) = 1 - P(E_1|D) - P(E_2|D) \\ = 1 - \frac{25}{69} - \frac{28}{69} = \frac{16}{69}$$

$E_1 \rightarrow$ Math graduate, $E_2 \rightarrow$ Non. math graduate

$P(A|E_1) = 0.30$, $P(A|E_2) = 0.05$. Hence $P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$
 After simplification we get $P(E_1|A)$ as desired.

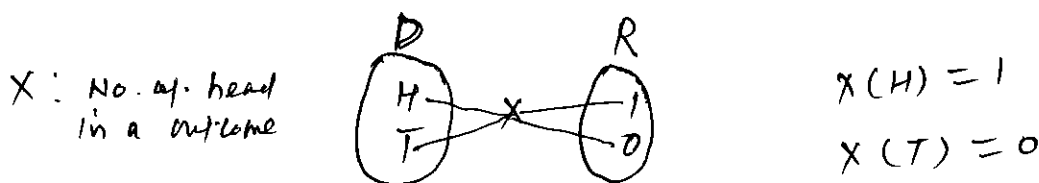
Ex. A bag contains 4 white and 2 black balls and second bag contains three of each colour. A bag is selected at random, and a ball is then drawn at random from the bag chosen. What is the prob that the ball drawn is white

$$\begin{aligned} P(\text{White}) &= P(\text{First bag chosen})P(\text{white is drawn}) \\ &\quad + P(\text{Second bag chosen})P(\text{white is drawn}) \\ &= \frac{1}{2} \cdot \frac{4}{6} + \frac{1}{2} \cdot \frac{3}{6} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12} \end{aligned}$$

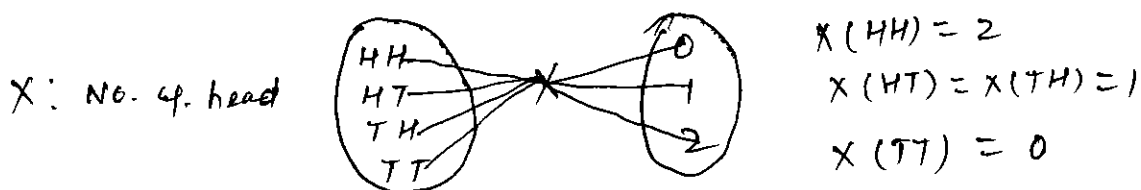
Random variable

A random variable is a real valued function associated with the outcomes of a random experiment.

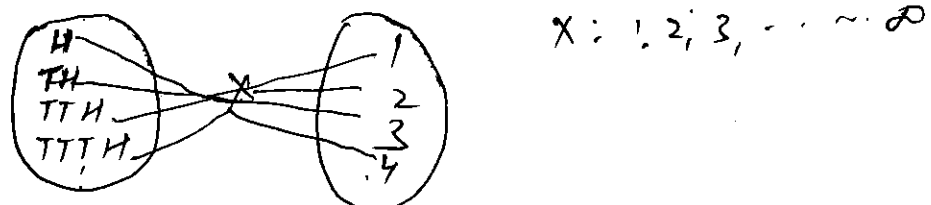
R.E. 1) Toss a coin possible outcomes H, T



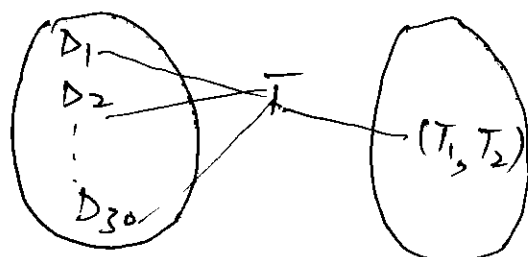
II) Toss two coins, possible outcomes HH, HT, TH, TT



III) No. of tosses required to get first time head if we toss a fair coin.



Temperature at particular place at given time at a month.



$$T_1 \leq T \leq T_2$$

The No. of possible values of r.v. may be discrete or continuous may be finite or infinitely large.

Prob Distⁿ : It is simply an arrangement to put the possible values of a random variable with their corresponding probabilities.

Ex. Toss a fair coin
Possible outcomes H, T
X : No. of head in a outcome

$$P(X(H)=1) = \frac{1}{2}, \quad P(X(T)=0) = \frac{1}{2}$$

X	0	1	
P(X)	$\frac{1}{2}$	$\frac{1}{2}$	1.00

$$P(X=x) = p(x)$$

Toss two ~~coins~~ coins

Possible outcome HH, HT, TH, TT

X : No. of head in a outcome.

$$X(HH)=2, \quad X(HT)=X(TH)=1, \quad X(TT)=0$$

$$P(X(HH)=2) = \frac{1}{4}, \quad P(X(HT)=1) = P(X(TH)=1) = \frac{2}{4}, \quad P(X(TT)=0) = \frac{1}{4}$$

X	0	1	2	
P(X)	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$	1.00

General symbol : $P(X=x) = p(x)$

Prob Mass Function (p.m.f.) (Defined for discrete r.v.)

If a discrete random variable X can have the values $x_1, x_2, \dots$ with probabilities $p(x_i), \dots$ ($i=1, 2, \dots$), then $p(x_i)$ is said to be p.m.f.

- iff
- i) $p(x_i) \geq 0 \quad \forall x_i \quad (i=1, 2, \dots)$
 - ii) $\sum_{i=1}^{\infty} p(x_i) = 1$

Prob Density Function (p.d.f.) (Defined for cont. r.v.)

If a Continuous r.v. X can assume all possible values in $(-\infty, \infty)$ with prob $f(x)$, then $f(x)$ is said to be p.d.f, iff

- i) $f(x) \geq 0 \quad \forall x \in (-\infty, \infty)$
- ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Cumulative distribution Function or Distribution Function (DF)

The distribution function of a random variable X is defined as

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} p(x_i) \quad \text{if } X \text{ is discrete r.v.}$$

$$= \int_{-\infty}^x f(x) dx \quad \text{if } X \text{ is continuous r.v.}$$

Ex .

x	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

$$F_X(1) = p(0) + p(1) = \frac{1}{4} + \frac{2}{4} = \frac{3}{4}$$

Mathematical Expectations

The mathematical expectation of a random variable X is defined as:

- i) For discrete r.v. X : If a discrete r.v. ~~can~~ ^{X} takes the values $x_1, x_2, \dots$ with probabilities $p(x_i)$ ($i=1, 2, \dots$) the mathematical expectation of the r.v. X is defined as

$$E(X) = x_1 p(x_1) + x_2 p(x_2) + \dots$$

$$= \sum_{i=1}^{\infty} x_i p(x_i) \quad ; \quad \text{provided the sum/series is convergent.}$$

ii) $1.e \quad \sum_{i=1}^{\infty} |x_i| p(x_i) < \infty$

For continuous r.v. X : If a continuous r.v. X takes all possible values in the interval $(-\infty, \infty)$ with prob $f(x)$, then the mathematical expectation of r.v. X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad , \quad \text{provided it exist.}$$

$$\text{provided } \int_{-\infty}^{\infty} |x| f(x) dx < \infty$$

Properties of Expectations

i) For a constant a , $E(aX) = aE(X)$

ii) For a constant a & b $E(aX + b) = aE(X) + b$
Constant is unaffected by expectation

iii) Addition Rule: For random variables $X_1, X_2, \dots, X_n$
 $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$

iv) Product Rule: For independent random variables $X_1, X_2, \dots, X_n$
 $E(X_1 X_2 \dots X_n) = E(X_1) E(X_2) \dots E(X_n)$

Moments in terms of Expectations

i) The r^{th} moment about origin is defined as

$$\mu_r' = E(X^r) ; \mu_0' = E(X^0) = E(1) = 1$$

$$\mu_1' = E(X) \rightarrow \text{Mean}, \mu_2' = E(X^2) \dots$$

ii) The r^{th} moment about a point A

$$\mu_r'(A) = E(X-A)^r$$

iii) The r^{th} central moment or moment about mean is defined as

$$\mu_r = E(X - E(X))^r = E(X - \mu_1')^r$$

$$\mu_0 = 1, \mu_1 = E(X - \mu_1') = E(X) - \mu_1' = \mu_1' - \mu_1' = 0$$

$$\mu_2 = E(X - E(X))^2 = E(X - \mu_1')^2 = \text{variance}$$

$$\mu_3 = E(X - E(X))^3, \mu_4 = E(X - E(X))^4 \text{ etc.}$$

iv) Relation between central moments and moments about origin.

$$\mu_2 = \mu_2' - \mu_1'^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

$$\beta_1 = \frac{\mu_3'^2}{\mu_2'^3}, \gamma_1 = \sqrt{\beta_1}$$

$$\beta_2 = \frac{\mu_4'}{\mu_2'^2}, \gamma_2 = \beta_2 - 3$$

Pearson's β & γ coefficients.

Moment Generating Function (MGF)

The m.g.f. of a random variable X is defined as

$$M_X(t) = E e^{tx} ; t \text{ is a real number.}$$

$$= \sum e^{tx} p(x) \text{ if } X \text{ is discrete r.v.}$$

$$= \int e^{tx} f(x) dx \text{ if } X \text{ is continuous r.v.}$$

Why it is called m.g.f? It generates moments.

$$M_X(t) = E e^{tX} = E \left[1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots \right]$$

$$= 1 + t h_1' + \frac{t^2}{2!} h_2' + \frac{t^3}{3!} h_3' + \dots + \frac{t^r}{r!} h_r' + \dots$$

$$\frac{\partial M_X(t)}{\partial t} = h_1' + \frac{2t}{2!} h_2' + \frac{3t^2}{3!} h_3' + \dots$$

$$\left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0} = h_1', \quad \text{Similarly we will get.}$$

$$\left. \frac{\partial^2 M_X(t)}{\partial t^2} \right|_{t=0} = h_2'$$

Proceeding in similar way in general we get

$$h_r' = \left. \frac{\partial^r M_X(t)}{\partial t^r} \right|_{t=0}$$

Properties of MGF

i) For a constant c

$$M_{cX}(t) = M_X(ct)$$

$$M_{cX}(t) = E e^{t(cX)} = E e^{t c X} = M_X(ct)$$

ii) $M_{X_1 + X_2 + \dots + X_n}(t) = M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t)$

(For independent r.v.s $X_1, X_2, \dots, X_n$)

$$M_{X_1 + X_2 + \dots + X_n}(t) = E e^{t(X_1 + X_2 + \dots + X_n)} = E(e^{tX_1} e^{tX_2} \dots e^{tX_n})$$

$$= E e^{tX_1} E e^{tX_2} \dots E e^{tX_n}$$

$$= M_{X_1}(t) M_{X_2}(t) \dots M_{X_n}(t) = \prod_{i=1}^n M_{X_i}(t)$$

iii) $M_U(t) = E e^{tU} = E e^{t \left(\frac{X-A}{h} \right)} = e^{-\frac{tA}{h}} E e^{\frac{t}{h} X} = e^{-\frac{tA}{h}} M\left(\frac{t}{h}\right)$

where $U = \frac{X-A}{h}$.